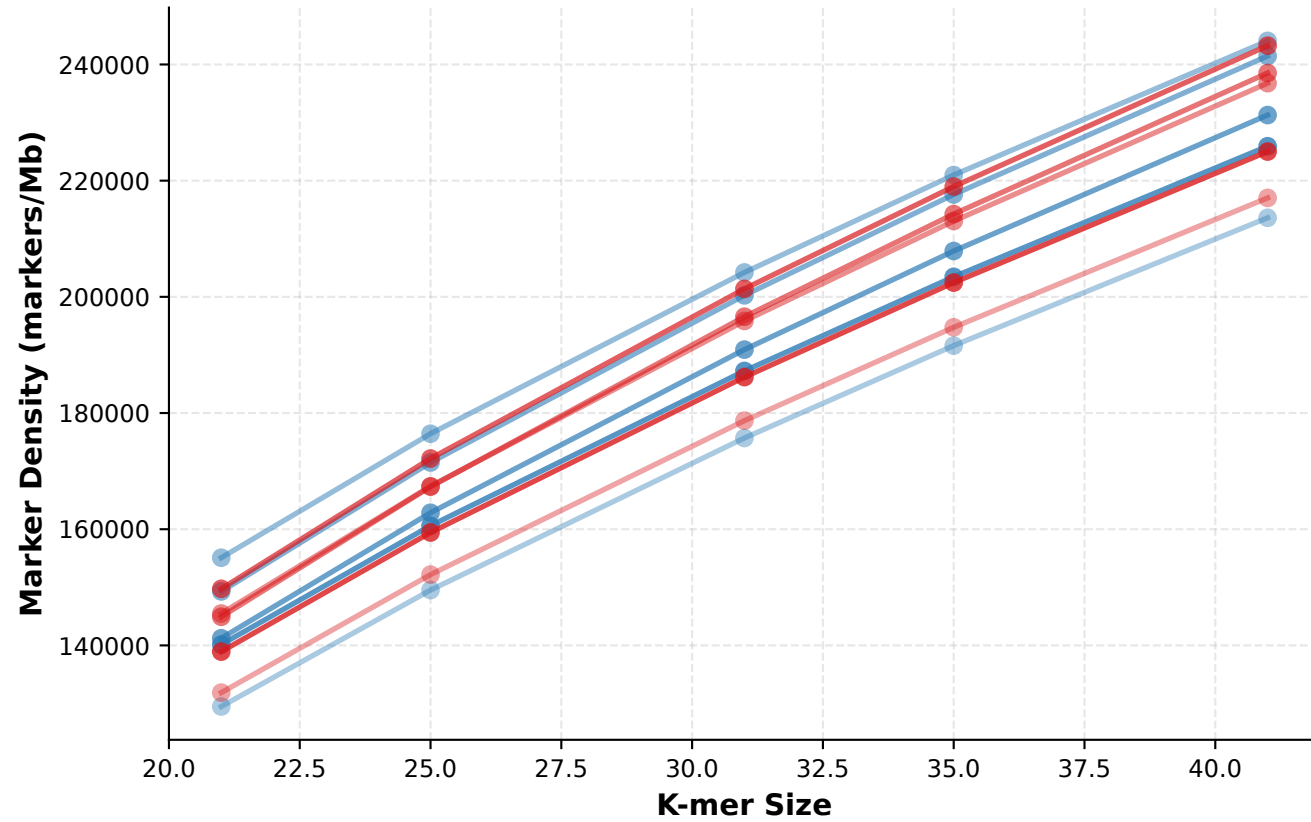
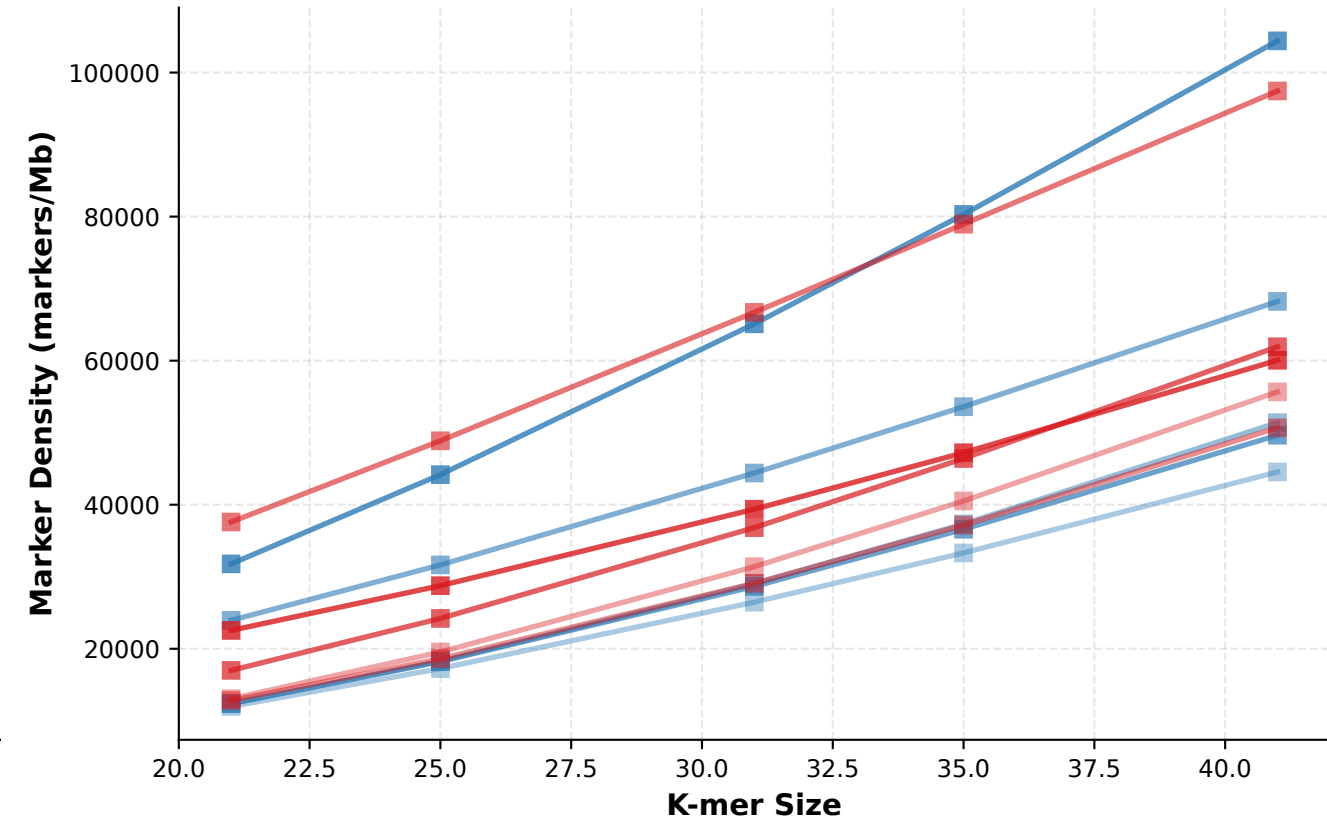


# Detailed K-mer Size Analysis by Genomic Region

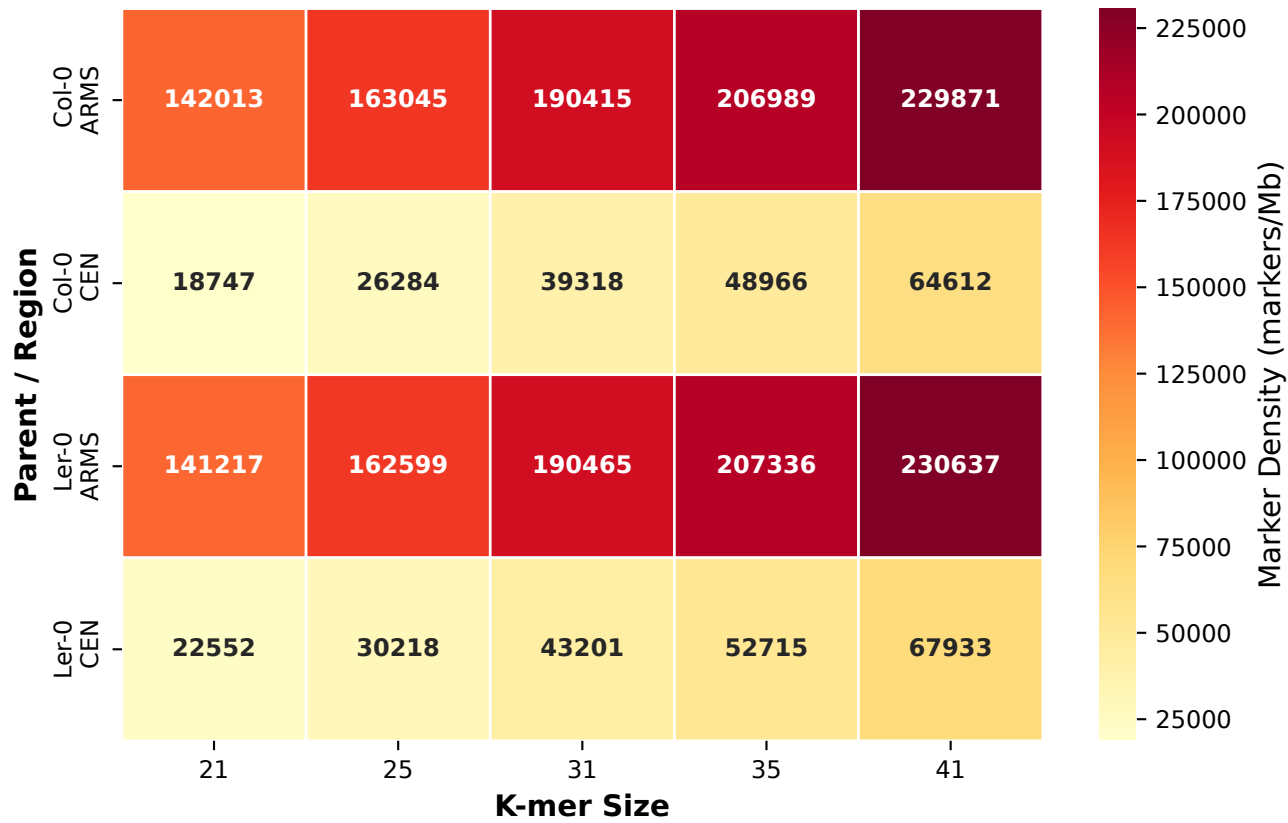
## ARMS Marker Density by Chromosome



## CEN Marker Density by Chromosome



## Marker Density Heatmap Across K-mer Sizes



### K-MER SIZE SELECTION GUIDE

Best parent balance: k=25  
(0.0% imbalance)

Most chromosome consistency: k=21  
(12.4% CV)

Maximum markers: k=41  
(57.9M markers)

#### RECOMMENDATIONS:

✓ All k-mer sizes are well-balanced

For high-quality reads:

→ Use k=41 (more markers)

For noisy reads:

→ Use k=21 (error tolerant)

Balanced choice:

→ Use k=31